Membrane Potential and Voltage-Gated Potassium Channels in Cancer: A Comprehensive Review and the Polaritin Concept

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Abstract

Cancer cells across diverse tissue types exhibit a conserved bioelectric signature: chronic membrane depolarization, with resting potentials typically ranging from −10 to −50 mV, compared with −60 to −90 mV in normal differentiated cells. This biophysical aberration is not merely correlative; depolarization promotes cell cycle progression, inhibits differentiation, and enhances migratory capacity, whereas experimental restoration of hyperpolarization reverses these malignant phenotypes. Despite compelling preclinical evidence, no clinically approved therapy directly targets cancer membrane potential, largely because existing ion channel modulators lack tumor selectivity and cause dose-limiting toxicity. This review synthesizes current understanding of membrane potential dysregulation in cancer, the role of voltage-gated potassium (Kv) channels as therapeutic actuators, and emerging strategies for bioelectric intervention. Building on this foundation, we propose Polaritin (极解素), a conceptual therapeutic framework employing a dual-gated (AND-logic) Kv channel modulator that activates exclusively in depolarized cancer cells only when an exogenous small-molecule ligand is present. We discuss the mechanistic rationale, computational design considerations, falsifiable predictions, and a staged validation roadmap. By anchoring selectivity in biophysical parameters intrinsic to cancer cells rather than mutable molecular biomarkers, this approach could theoretically retain efficacy across genetically heterogeneous tumors.

Keywords: cancer bioelectricity; membrane potential; voltage-gated potassium channels; Kv10.1; Kv1.3; de novo protein design; targeted therapy; depolarization; dual-gating; AND-logic

1. Introduction

The somatic mutation theory of cancer has dominated oncology for decades, yielding targeted therapies and immunotherapies that have transformed outcomes for many malignancies. Yet a substantial fraction of tumors remains refractory, and acquired resistance through mutational escape continues to undermine even the most sophisticated molecularly targeted agents [1,2]. This persistent challenge has motivated exploration of non-genetic layers of cancer regulation—metabolic, epigenetic, and biophysical—that may represent more stable therapeutic vulnerabilities.

Among these non-genetic dimensions, bioelectric signaling has emerged as a particularly promising frontier. Since the pioneering observations of Burr in the 1930s and the theoretical work of Cone in 1971 [3], accumulating evidence has established that cancer cells possess a distinct electrophysiological identity characterized by chronic plasma membrane depolarization [4,5]. This depolarized state actively drives proliferation, suppresses differentiation, and promotes invasion—three cardinal features of cancer [6,7]. Crucially, experimental manipulation of membrane potential toward hyperpolarization can reverse these phenotypes [8,9].

Despite this robust evidence base, the bioelectric dimension of cancer remains therapeutically untapped in the clinic. The primary obstacle is selectivity: ion channels are expressed ubiquitously, and systemically administered channel modulators invariably cause dose-limiting cardiovascular and neurological toxicity [10]. In this review, we synthesize the current literature on cancer bioelectricity, with particular focus on voltage-gated potassium (Kv) channels as potential therapeutic actuators. We then present Polaritin (极解素), a conceptual framework employing a dual-gated Kv channel modulator designed to activate exclusively in depolarized cancer cells.

2. Membrane Potential as a Conserved Bioelectric Hallmark of Cancer

2.1 The depolarized phenotype of cancer cells

A defining electrophysiological feature of malignant cells is their chronically depolarized resting membrane potential (V_m). Across numerous cancer types, tumor cells consistently exhibit V_m values of approximately −10 to −50 mV, whereas normal differentiated somatic cells typically maintain −60 to −90 mV [4,5,11]. The discriminant

threshold between proliferative and terminally differentiated cells has been estimated at approximately −36 mV [12]. This depolarization is driven by altered expression of ion channels and transporters, including upregulated voltage-gated sodium channels, reduced inward-rectifier potassium channels, and modulated chloride homeostasis [13,14].

2.2 Functional consequences of depolarization

Depolarization promotes malignancy through multiple mechanisms: voltage-dependent calcium influx activates proliferative signaling (calcineurin/NFAT, MAPK) [15,16]; sustained depolarization maintains a stem-like, undifferentiated state [17,18]; depolarization enhances migration through cytoskeletal and volume regulation [19,20]; and depolarized potentials associate with apoptosis resistance [21]. Conversely, restoration of hyperpolarization—through potassium channel openers, optogenetic tools, or ion channel overexpression—reduces proliferation, induces differentiation, and triggers apoptosis in preclinical models [8,9,22].

3. Ion Channels in Cancer: Mechanisms and Therapeutic Relevance

Ion channels are increasingly recognized as key regulators of cancer hallmarks, termed "oncochannelopathies" [23]. Voltage-gated sodium channels (e.g., Nav1.5) enhance invasiveness in metastatic cancers but cause cardiac and neurological toxicity when blocked [24,25]. Calcium-permeable channels (ORAI, TRP) regulate proliferation and migration [26], while chloride channels (CLIC1, VRAC) control cell volume during migration [27]. Potassium channels constitute the largest and most diverse family, with over 80 human genes, and several subtypes—Kv1.3, Kv10.1, Kv11.1, KCa3.1—are specifically upregulated in cancers [10,28,29].

4. Voltage-Gated Potassium Channels: From Biology to Therapeutic Targeting

4.1 Kv1.3 and Kv10.1: key oncogenic channels

Kv1.3 (KCNA3), with an activation midpoint of approximately −38 mV, is expressed in lymphocytes and upregulated in breast, colon, and pancreatic cancers [30,31]. Its activation threshold closely overlaps the depolarized range of cancer membranes. Kv10.1 (KCNH1, EAG1) is normally expressed almost exclusively in the

central nervous system but is aberrantly expressed in over 70% of human tumors [32,33]. It features a large extracellular S3–S4 linker and a distinctive voltage-sensing mechanism with six positively charged S4 residues [34,35]. Its tumor-restricted expression makes Kv10.1 particularly attractive, as normal tissue expression is largely protected by the blood-brain barrier.

4.2 Existing therapeutic approaches and limitations

Strategies targeting potassium channels in cancer include small-molecule inhibitors, peptide toxins, monoclonal antibodies, and antisense oligonucleotides [10,36]. A monoclonal antibody against Kv10.1 (mAb56) demonstrated anti-tumor efficacy in xenografts [37]. However, common limitations persist: incomplete tumor selectivity, need for continuous administration, compensatory channel upregulation, and inability to exploit the full membrane potential differential between cancer and normal cells. These limitations motivate more sophisticated, logic-gated therapeutic strategies.

5. The Polaritin Framework: Dual-Gated Selective Repolarization

5.1 Core concept: biophysical selectivity through AND-logic gating

The central innovation of the Polaritin framework is the proposal that tumor selectivity can be achieved through a logical AND operation requiring two independent conditions: (1) membrane depolarization above a defined threshold (~−35 mV), and (2) binding of an exogenous small-molecule ligand absent from the physiological milieu. Only when both conditions are satisfied would the engineered Kv channel modulator open, driving K^+ efflux and repolarization toward $E_k \approx -90$ mV (Figure 1). This dual-gate architecture is conceptually analogous to chemogenetic DREADD systems and split-protein biosensors [38,39], adapted for electroceutical application.

The AND-logic truth table (Supplementary Figure 1) illustrates the four possible input combinations: only the depolarized-plus-ligand condition produces channel opening and K^+ efflux. Normal cells at −70 mV remain unaffected even with ligand present; depolarized cancer cells without ligand also remain silent. This multiplicative selectivity —voltage gate × ligand gate—provides a therapeutic window unattainable by either mechanism alone.

5.2 The first gate: voltage-threshold selectivity

The voltage gate leverages the intrinsic voltage sensitivity of Kv channel voltage-sensing domains (VSDs). By engineering the VSD activation threshold to approximately −35 mV—close to the discriminant potential between proliferative and differentiated cells—the channel responds preferentially to sustained cancer depolarization. Mathematical modeling predicts that a modest 2–5 nS/pF increase in Kv conductance suffices to repolarize a typical cancer cell by 15–25 mV [40], a magnitude associated with anti-proliferative effects [6,7]. Transient physiological depolarization in normal cells (e.g., immune activation) is brief and may not sustain significant channel activation.

5.3 The second gate: exogenous ligand control

The ligand gate provides additional selectivity and external controllability. The channel remains closed regardless of membrane potential unless the exogenous ligand is present. Because the ligand is administered systemically but the channel only activates in depolarized cells, the effective therapeutic window is widened by the product of two selectivities. Channel activity can be rapidly terminated by discontinuing ligand administration, providing a pharmacological safety switch. The ligand-binding module could leverage validated chemogenetic scaffolds (e.g., DREADD-type systems responsive to clozapine-N-oxide or deschloroclozapine) [38,41], or natural product ligands such as flavonoids (apigenin, luteolin) with demonstrated Kv channel modulatory activity and excellent safety profiles [42]. The choice should prioritize orthogonality to endogenous physiology, favorable pharmacokinetics, and established human safety data.

6. Computational Design and Structural Considerations

6.1 AI-driven de novo protein design pipeline

Recent advances in AI-driven protein structure prediction and design—AlphaFold 3 [43], RFdiffusion [44], and ProteinMPNN [45]—have dramatically lowered the barrier to designing novel proteins with predefined functions. The proposed design pipeline (Supplementary Figure 2) proceeds as follows: (1) structure prediction of the target Kv channel VSD in depolarized and resting conformations using AlphaFold 3 / ColabFold [46]; (2) generation of novel binder scaffolds via RFdiffusion, specifically recognizing

the depolarized-state VSD conformation; (3) sequence design of the binder interface using ProteinMPNN; (4) iterative validation of folding and binding specificity through AlphaFold 3 complex prediction; and (5) molecular dynamics simulation to assess binding stability and allosteric effects on channel gating.

6.2 Structural targeting of the Kv10.1 voltage-sensing domain

As a primary target, the Kv10.1 VSD offers tumor-restricted expression, a large extracellular S3–S4 linker (residues 310–345 in human KCNH1) accessible to protein binders, and a well-characterized voltage-sensing mechanism [34,35]. The S3–S4 linker is enriched in acidic residues (E314, D316, E318, D322, D325, E339), creating a negatively charged extracellular surface complemented by a positively charged binder interface. The S4 helix, with conserved positively charged residues (K354, R357, R360, R363, R366 in human KCNH1), undergoes conformational displacement during voltage sensing; a binder stabilizing the activated S4 conformation would promote channel opening (Supplementary Figure 4).

A preliminary design concept for the Polaritin binder envisions a compact three-helix bundle protein of approximately 88–100 residues with a net positive charge at physiological pH, enabling electrostatic complementarity with the acidic S3–S4 linker. The third helix presents the primary binding interface, with mixed charged and hydrophobic residues tailored to engage both the S3–S4 linker and the extracellular N-terminus of S4. Binding of Polaritin to the depolarized-state VSD allosterically stabilizes the activated S4 conformation, lowering the activation threshold and promoting K^+ efflux. The exogenous ligand acts as an additional allosteric modulator, potentially binding at the intracellular pore domain (S6 helix central cavity) to stabilize the open state, creating positive cooperative dual-gating.

6.3 Molecular dynamics and binding free energy validation

Computational validation employs molecular dynamics (MD) simulations using GROMACS, AMBER, or OpenMM with explicit solvent. Simulation of the Kv channel–Polaritin–ligand ternary complex over 50–100 ns assesses binding stability, conformational dynamics, and allosteric communication. Binding free energies calculated

via MM/PBSA or MM/GBSA quantify interaction strength and specificity, while per-residue decomposition identifies key contact residues for experimental validation [47].

7. Falsifiable Predictions and Validation Roadmap

7.1 Three core predictions

The dual-gate design generates three falsifiable predictions: (1) Cells maintained at −35 mV will respond to the dual-gated modulator with measurable hyperpolarization and reduced proliferation, whereas cells clamped at −70 mV will not, even with ligand. (2) Ligand alone, without depolarization, produces no electrophysiological response. (3) In co-culture of cancer and normal cells, selective repolarization of the cancer population is achievable at ligand concentrations that do not affect normal cells.

7.2 Staged validation roadmap

Validation proceeds through five stages: Stage 1—Computational design and validation (AlphaFold 3, RFdiffusion, molecular docking, MD). Stage 2—Protein production and biophysical characterization (recombinant expression, circular dichroism, surface plasmon resonance). Stage 3—Electrophysiological characterization in heterologous systems (patch-clamp recording). Stage 4—Selectivity testing in mixed-cell assays (cancer/normal co-culture). Stage 5—Efficacy in tumor spheroid and xenograft models.

8. Limitations and Challenges

Several challenges warrant acknowledgment: (1) cancer membrane potential heterogeneity may render subpopulations unresponsive; (2) chronic Kv activation may trigger compensatory conductance changes; (3) grafting ligand sensitivity onto Kv pores without disrupting voltage sensing remains unproven; (4) immunogenicity of de novo designed proteins is a clinical concern; (5) protein delivery to tumors presents pharmacokinetic challenges; and (6) long-term effects of sustained repolarization on excitable tissues require careful evaluation.

9. Conclusions and Future Perspectives

The bioelectric dimension of cancer represents a therapeutically untapped vulnerability distinct from genetic biomarkers. The Polaritin framework—employing a dual-gated Kv channel modulator with AND-logic activation—offers a conceptual strategy for converting cancer's characteristic depolarization into a selective apoptotic signal while sparing normal tissue (Supplementary Figure 3). By anchoring selectivity in biophysical parameters intrinsic to cancer cells rather than mutable molecular markers, this approach could theoretically retain efficacy across genetically heterogeneous tumors. The convergence of AI-driven protein design, ion channel structural biology, and cancer bioelectricity creates an unprecedented opportunity to translate bioelectric concepts into therapeutic reality.

Figure Legends

Figure 1. Conceptual schematic of the Polaritin dual-gate (AND-logic) activation mechanism. (a) Normal polarized cell at −70 mV: even in the presence of the exogenous ligand, the channel remains closed because the membrane potential is far below the activation threshold. (b) Depolarized cancer cell at −35 mV without ligand: the voltage gate is satisfied, but the absence of the ligand keeps the channel closed. (c) Depolarized cancer cell at −35 mV with ligand bound: both gates are satisfied simultaneously, the channel opens, and sustained K^+ efflux drives repolarization toward $E_k \approx -90$ mV, engaging endogenous apoptotic pathways. The channel opens only when both conditions are met, providing tumor-selective and externally controllable activity.

Supplementary Figure 1. Dual-gate AND-logic truth table. Four possible input combinations of membrane depolarization (above/below −35 mV threshold) and exogenous ligand (present/absent) are enumerated. Only the depolarized-plus-ligand condition produces channel opening, K^+ efflux, and apoptosis. All other three combinations result in channel closure and cell survival, demonstrating the multiplicative selectivity of the AND-logic design.

Supplementary Figure 2. AI-driven de novo protein design pipeline. Sequential workflow integrating target structure determination (KV10.1 VSD), RFdiffusion backbone generation, ProteinMPNN sequence design, AlphaFold 3 structural validation, and molecular dynamics / MM-PBSA binding free energy calculation. Each rejected design feeds back constraints into the next generation cycle, converging on 10–20 high-confidence binder candidates.

Supplementary Figure 3. Therapeutic paradigm comparison. Top: three conventional modalities (chemotherapy, targeted therapy, immunotherapy) with their respective limitations. Bottom: Polaritin electrophysiological therapy targets cancer-cell membrane depolarization, a universal bioelectric hallmark, offering mutation-independent broad-spectrum activity, a dual safety switch with >100-fold therapeutic window, and physical K^+ efflux triggering intrinsic apoptosis.

Supplementary Figure 4. Polaritin binding interface with the KV10.1 voltage-sensing domain. Schematic showing the Polaritin three-helix bundle (orange) engaging the extracellular S3–S4 linker and S4 helix of the KV10.1 VSD (blue). Salt bridges form between Polaritin's positively charged residues (K/R) and the acidic S3–S4 linker (E/D). Polaritin binding allosterically pushes the S4 helix outward, stabilizing the activated conformation and promoting K^+ efflux and repolarization.

Figure Preparation

Schematic diagrams (Figure 1 and Supplementary Figures 1–S4) were constructed using Python-based vector graphics rendering (Matplotlib). No generative-AI image-generation tools were used in the preparation of any figure.

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